AN ELEMENT OF A VALUATION RING IS A NONZERODIVISOR ON A TORSION FREE MODULE EXACTLY WHEN THE MODULE IS FLAT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Flatness of a module over a ring tested by the equational criterion, meaning every relation is a combination of trivial relations. the author cites Algebra, Lemma algebra-lemma-flat-eq at proof lines 4932-4934. Supplies the criterion that the proof verifies.
- Valuation ring and its valuation, which is totally ordered so that a finite set of elements has one of minimal valuation. used at proof lines 4936-4940. Allows the coefficients of the relation to be factored through one of them.
- A flat module over a domain is torsion free, used here for the easy direction. the statement counted by the sibling card S1-algebra-flat-torsion-free at 4913-4922, cited at proof lines 4930-4931. Gives the first implication.

Source reference. Original lemma 0539: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6597–6597. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0538: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6596–6596. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 646–646. Navigation pointer to a statement the

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Lemma 0.2. *Let A be a valuation ring. An A-module M is flat over A if and only if M is torsion free.*

Proof. The implication "flat $\Rightarrow$ torsion free" is Lemma 0538. For the converse, assume M is torsion free. By the equational criterion of flatness (see Algebra, Lemma 00HK) we have to show that every relation in M is trivial. To do this assume that $\sum_{i=1,\ldots,n} a_i x_i = 0$ with $x_i \in M$ and $a_i \in A$. After renumbering we may assume that $v(a_1) \leq v(a_i)$ for all i. Hence we can write $a_i = a_i' a_1$ for some $a_i' \in A$. Note that $a_1' = 1$. As M is torsion free we see that $x_1 = -\sum_{i \geq 2} a_i' x_i$. Thus, if we choose $y_i = x_i$, $i = 2, \ldots, n$ then

$$x_1 = \sum\nolimits_{j \geq 2} -a_j' y_j, \quad x_i = y_i, (i \geq 2) \quad 0 = a_1 \cdot (-a_j') + a_j \cdot 1 (j \geq 2)$$

shows that the relation was trivial (to be explicit the elements a_{ij} are defined by setting $a_{11} = 0$, $a_{1j} = -a_j'$ for $j > 1$, and $a_{ij} = \delta_{ij}$ for $i, j \geq 2$). $\square$